

Report: MA6217 Problem-Based Assignment 1

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Check all the codes, Simulink files, images, raw data and report at ntulearn or:
github.com/MrOxOrOxen/ntu/tree/ma6217/PBA1.

1 Task 1

1.1 Overall system architecture

Figure 1 shows the architecture of the overall top-level system in Simulink.

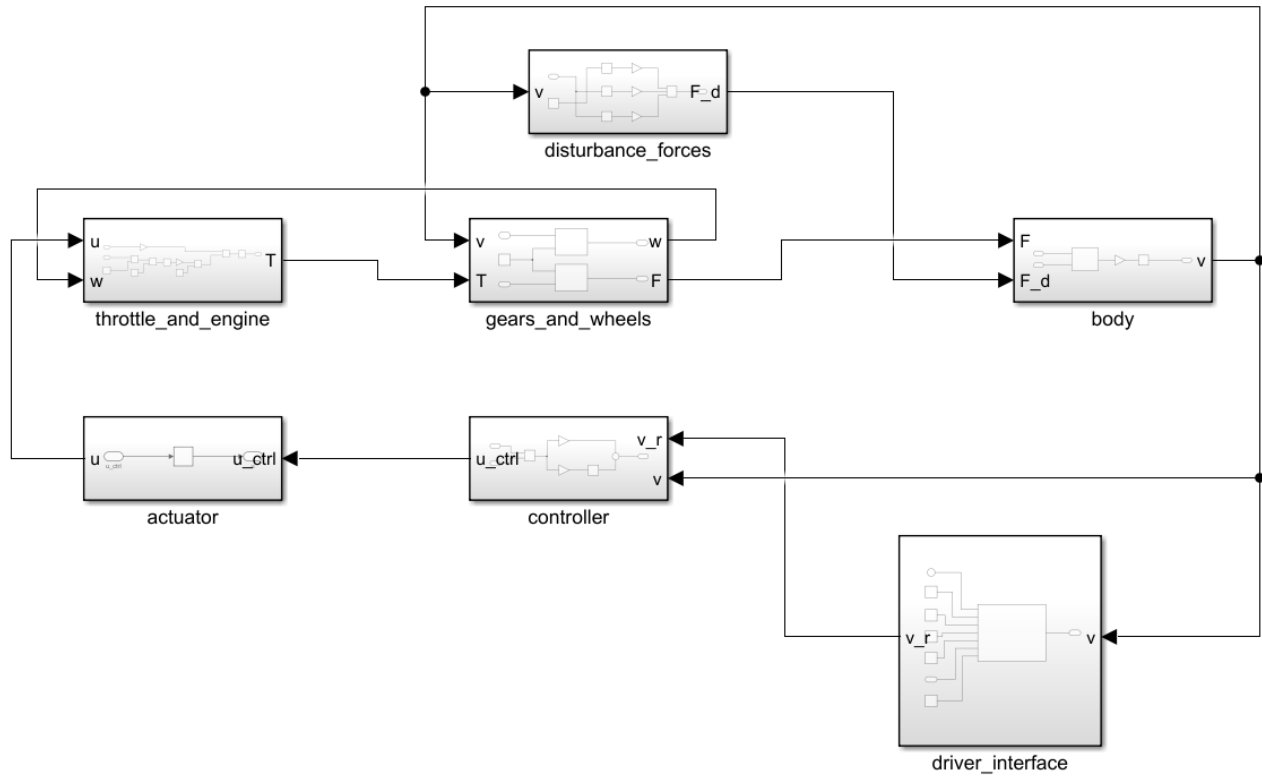


Figure 1: Overall system architecture

Table 1 shows the meanings of the physical quantities shown in figure 1.

As illustrated in Figure 1, the overall model operates as a closed-loop control system. The overall signal paths flow as follows:

- *Driver interface*: Generate the target speed v_r from the user input (cruise control signal) and the real-time vehicle speed v .

Physical Quantity	Meaning	Unit
u_{ctrl}	Ideal throttle signal	/
u	Actual throttle position	/
ω	Engine rotational speed	rad/s
T	Engine torque	$N \cdot m$
v	Actual vehicle speed	m/s
v_r	Target vehicle speed	m/s
F	Engine traction force	N
F_d	Disturbance force	N

Table 1: Physical Quantities in overall structure

- *Controller*: Input target speed v_r and real-time speed v , compute tracking error, and generate the control signal u_{ctrl} .
- *Actuator*: Translate u_{ctrl} into the actual throttle position u .
- *Throttle and Engine*: Input the throttle position u and engine rotational speed ω , and calculate engine torque T .
- *Gears and wheels*: Receive engine torque T and vehicle speed v , and convert them to traction force F while back-calculating the engine rotational speed ω based on the selected gear ratio α_n .
- *Disturbance forces*: Calculate external resistance forces F_d based on the vehicle speed v .
- *Body*: Input the traction force F and disturbance force F_d , and computes the acceleration and integrates it to vehicle speed v .

The details of the subsystems are as follows.

1.2 Driver Interface

Figure 2 shows the structure of the subsystem *Driver interface*, where *clock* is the simulation time (will be used in the following tasks), and the other inputs are loaded from the file *constants.m*.

The *driver* MATLAB function models driver interaction with the cruise control system, in which persistent variables *target_v* and *is_active* maintain system state memory access execution time steps. The subsystem logic handles five distinct operational conditions:

1. System disablement (*on_off* = 0): Deactivates cruise control (*is_active* = *false*) and directly passes through the current vehicle speed v to v_r .
2. Cruise cancellation (*cancel* = 1): The same as above, but can store the current speed v in the variable *target_v* for future resumption.
3. Target speed set (*set_decel* = 1): Activates cruise mode (*is_active* = *true*) and latches *ideal_v_r* as the new setpoint velocity *target_v*.
4. Speed increment (*resume_accel* = 1):
 - If cruise control is active, it increase the current setpoint by 1 m/s in each sampling time (i.e. 0.01 s in my setting).
 - If inactive, it re-activates the cruise at *target_v*.
5. Output multiplexing: When *is_active* is *true*, the function outputs the latched *target_v*. Otherwise, it outputs the actual vehicle speed v .

The conditions mean that if cruise is activated, the goal speed will be the speed that the driver sets manually; otherwise, the goal speed will be the actual speed of the vehicle.

Set the sampling time of MATLAB function as 0.01 s rather than -1 to prevent runtime error.

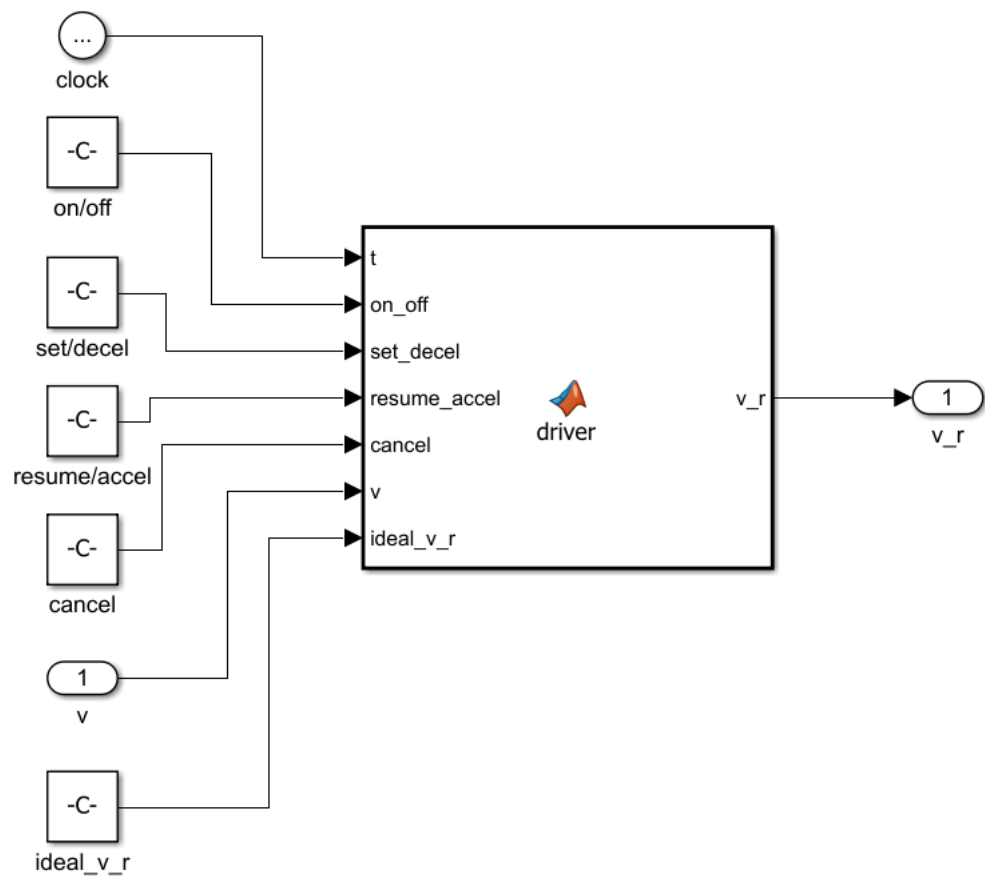


Figure 2: Structure of Subsystem *Driver Interface*

1.3 Controller

Figure 3 shows the structure of the subsystem *Controller*, where k_i and k_p are self-defined in the file *constants.m*.

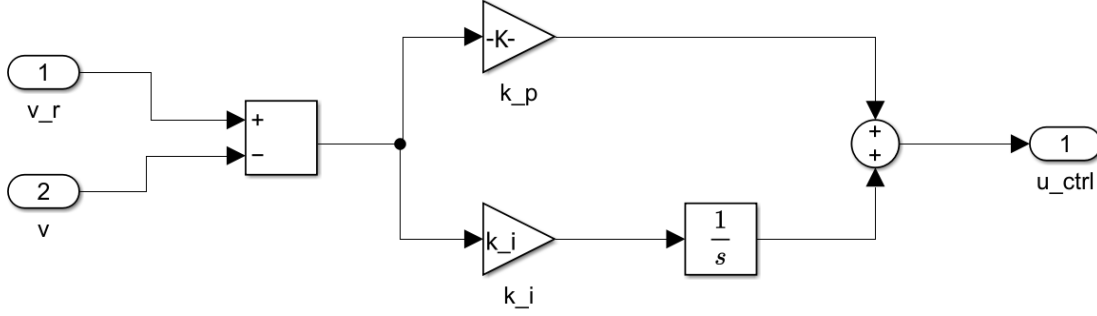


Figure 3: Structure of Subsystem *Controller*

The controller follows the PI control equation:

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau \quad (1)$$

where $e(t) = v_r - v$ is the system tracking error. Initial conditions of the integrator vary according to the requirements of different tasks.

1.4 Actuator

Figure 4 shows the structure of the subsystem *Actuator*. The saturation block is used to limit the maximum and minimum value of the input u_{ctrl} , i.e.

$$u(t) = \begin{cases} 1, & u_{ctrl}(t) > 1 \\ 0, & u_{ctrl}(t) < 0 \\ u_{ctrl}(t), & else \end{cases} \quad (2)$$

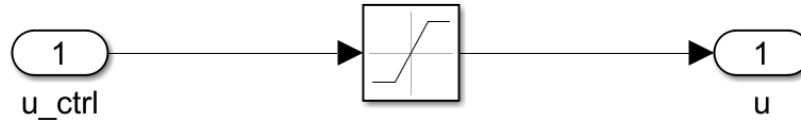


Figure 4: Structure of Subsystem *Actuator*

For most of the times, u is larger than 1, so the output is mostly 1.

1.5 Throttle and engine

Figure 5 shows the structure of the subsystem *Throttle and engine*, where:

- The *gain* block after the input u has a gain T_m .
- The *gain* block after the square block has a gain β .
- The variables ω_m, β and T_m come from the file *constants.m*.

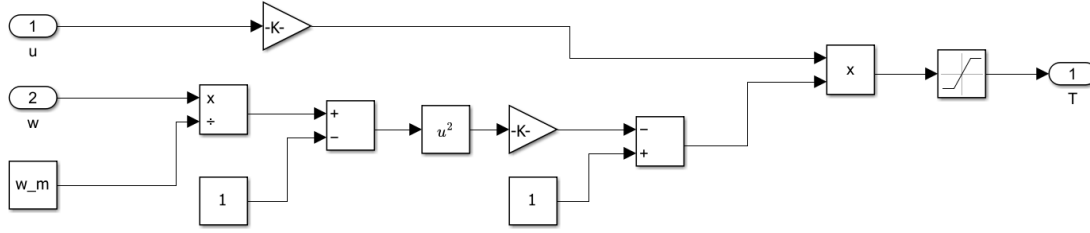


Figure 5: Structure of Subsystem *Throttle and engine*

There is a saturation block before the output T to make sure $T \geq 0$.

The subsystem follows the following equation:

$$T(\omega, u) = u \cdot T_m [1 - \beta (\frac{\omega}{\omega_m} - 1)^2] \quad (3)$$

1.6 Gears and wheels

Figure 6 shows the structure of the subsystem *Gears and wheels*, where the constant block *alpha for task 1-3* is the variable value of α_1 to α_5 in the file *constants.m*, and the constant block will be replaced by a stateflow chart in task 4.

Change the value of the constant block to simulate the variable gear selection ratios α_n in tasks 1, 2 and 3.

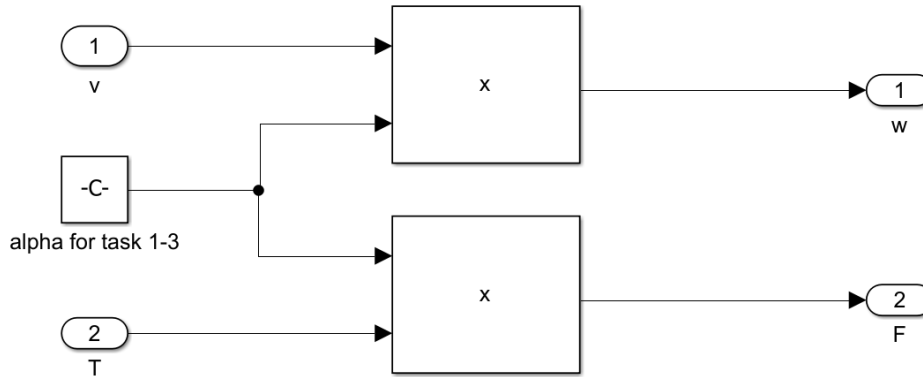


Figure 6: Structure of Subsystem *Gears and wheels*

The subsystem follows the following equations:

$$\omega = \alpha_n v \quad (4)$$

$$F = \alpha_n T \quad (5)$$

where $n \in \{1, 2, 3, 4, 5\}$.

1.7 Disturbance forces

Figure 7 shows the structure of the subsystem *Disturbance forces*, where:

- The *gain* block after the sine block has a gain of $m \cdot g$.
- The *gain* block after the sign block has a gain of $m \cdot g \cdot C_r$.

- The *gain* block after the square block has a gain of $\frac{1}{2}\rho \cdot C_d \cdot A$.
- The variables $\theta, m, g, C_r, C_d, \rho$ and A come from the file *constants.m*.

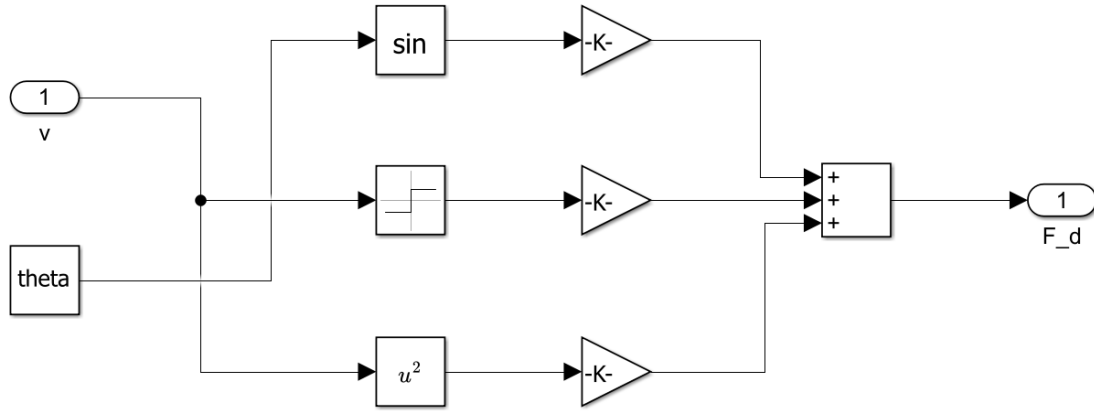


Figure 7: Structure of Subsystem *Disturbance forces*

The subsystem follows the following equation:

$$F_d = mg \sin \theta + mg C_r \operatorname{sgn}(v) + \frac{1}{2} \rho C_d A v^2 \quad (6)$$

1.8 Body

Figure 8 shows the structure of the subsystem *Body*, where the *gain* block after the minus block has a gain of $\frac{1}{m}$, and m comes from the file *constants.m*.

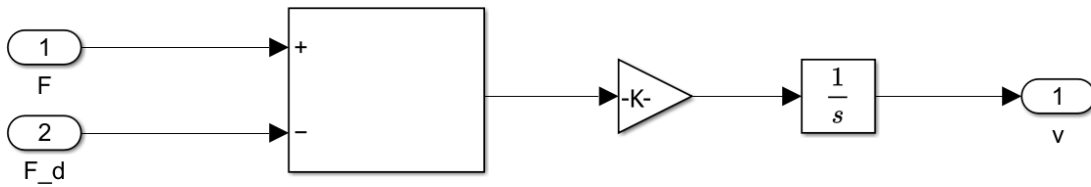


Figure 8: Structure of Subsystem *Body*

The subsystem follows the following equation:

$$m \frac{dv}{dt} = F - F_d \quad (7)$$

i.e.

$$v(t) = \int_0^t \frac{dv}{d\tau} d\tau = \int_0^t \frac{1}{m} [F(\tau) - F_d(\tau)] \quad (8)$$

Initial conditions of the integrator vary according to the requirements of different tasks.

2 Task 2

Since task 2 is open-loop and owns a constant value of input u , we cut the u connection of subsystem *Actuator* and *Throttle and engine*, and add a constant block to represent u instead. Figure 9 shows the new overall figure, where the subsystem *Throttle and engine* receives the constant input u .

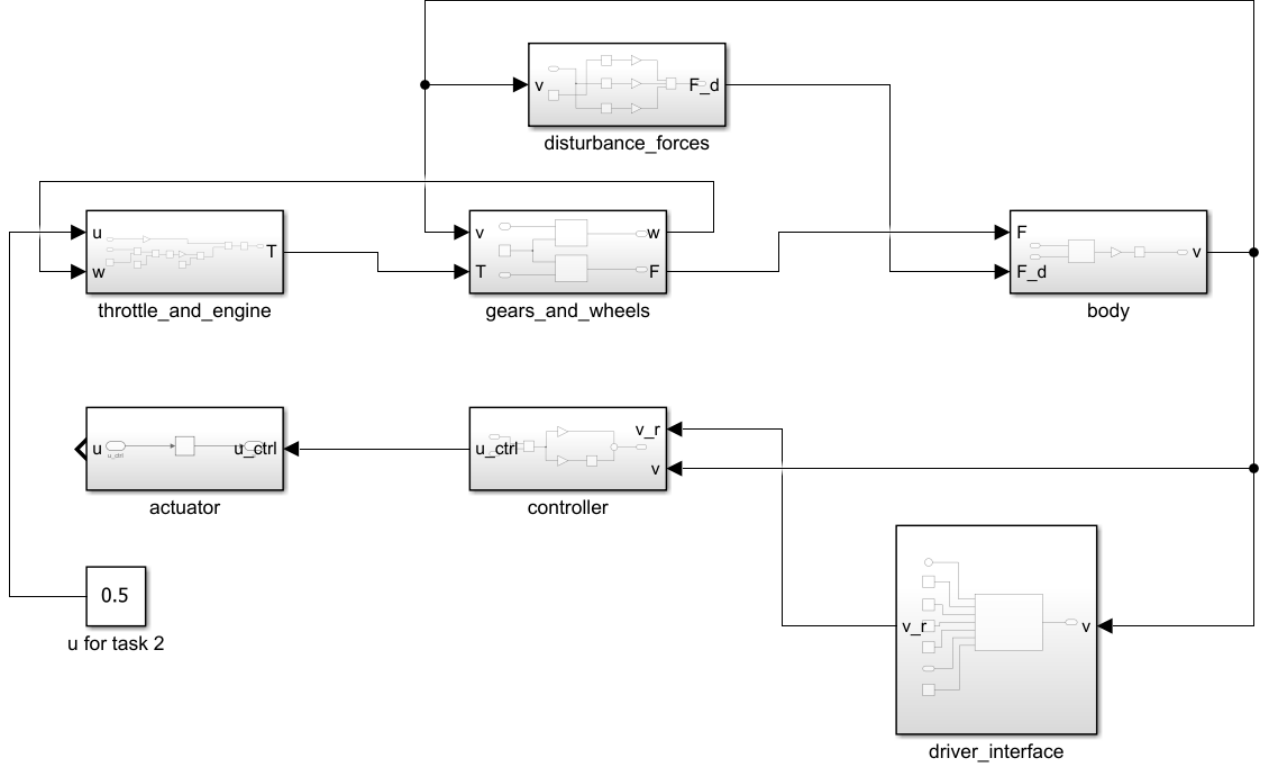


Figure 9: Overall system architecture in Task 2

2.1 Simulation Conditions

The initial condition of the integrator (which is defined as the output of the integrator in Simulink):

- In subsystem *Body*: 0, since the initial velocity of the vehicle is 0.
- In subsystem *Controller*: Any value, since the whole system is open-loop and the *Controller* block is after the output v . The initial value of the integrator does not affect the output v .

Similarly, the initial value of the self-defined variables (k_p, k_i, v_r) doesn't affect the output value v . We can set them as any values.

Hence, Table 2 shows the simulation conditions as follows.

2.2 Result

Figure 10 shows the velocity-time image of the open-loop control in task 2.

It is clear that the vehicle accelerates from the start, but stables at a certain velocity after a period of time, since the disturbance force increases with the increase of speed v according to equation 6, and according to equation 7 and 8, the acceleration will be 0 because $F - F_d = 0$, and there is no velocity increase then.

Actually, I set the simulation time as 100 s, and Figure 10 only shows the result in the first 40 seconds. According to the raw data, the output v is stable when the time approaches 100 s.

Parameter	Variable	Value	Unit
Gear	/	1	/
Transmission ratio	$\alpha_n(\alpha_1)$	40	rad/m
Throttle position	u	0.5	/
Road slope	θ	0	deg
Initial speed (integrator initial condition)	$v(0)$	0	m/s
Goal speed	v_r	any	m/s
Proportion Coefficient	k_p	any	/
Integral Coefficient	k_i	any	/
Output	v	/	m/s

Table 2: Simulation Conditions in Task 2

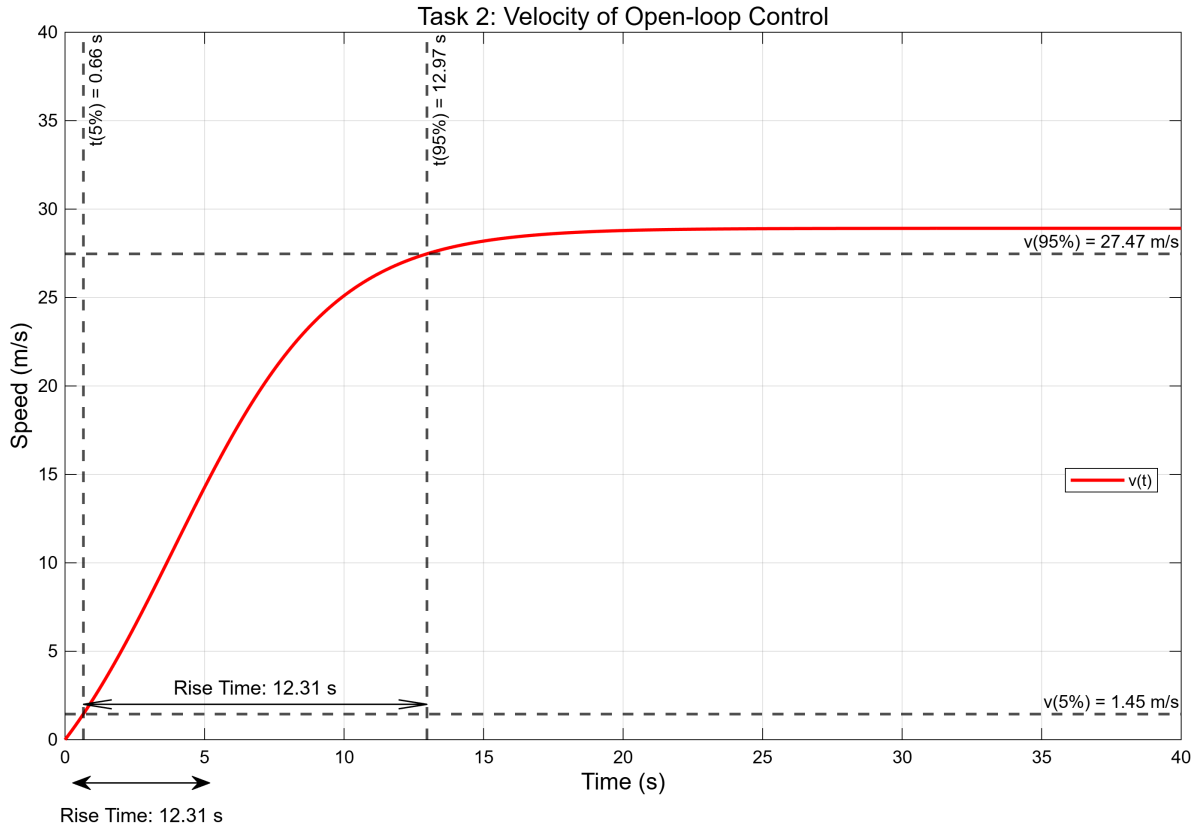


Figure 10: Velocity of open-loop control in Task 2

Hence, we can consider that the mean speed over the final 5% of the simulation (i.e. 95 - 100 s) can be regarded as the steady-state speed. And by the way, by calculating the mean speed of the last 5 seconds, we can define how accurately the final speed matches the target speed.

According to the raw data and code execution result, the simulation results are as follows:

Initial speed	0 m/s
Vehicle speed at the end of t	28.9151 m/s
Steady-state speed	28.9151 m/s
5% of steady-state speed	1.4458 m/s
95% of steady-state speed	27.4693 m/s
Time for 5% speed	0.66 s
Time for 95 % speed	12.97 s
Rise time	12.31 s

Table 3: Simulation Result of Task 2

3 Task 3

Task 3 is a closed-loop task, so its top-level simulation architecture is the same as Figure 1, but with a different input $\alpha_3 = 16$.

3.1 Simulation Conditions

The initial condition of the integrator (the output of the integrator at $t = 0^+$):

- In subsystem *Body*: 20, since the initial velocity of the vehicle is 20 m/s.
- In subsystem *Controller*: 0.1042, whose steps of calculating is as follows.

Since task 3 asks us to maintain the vehicle velocity v at 20 m/s for a period of time, we need to ensure the acceleration a of the vehicle at $v = 20\text{m/s}$ is 0, which is:

$$F - F_d = 0 \quad (9)$$

$$F = F_d = mg \sin \theta + mg C_r \text{sgn}(v) + \frac{1}{2} \rho C_d A v^2 = 323.64 N \cdot m \quad (10)$$

$$T = \frac{F}{\alpha_3} \approx 20.2275 N \cdot m \quad (11)$$

And then reverse calculate the throttle position u to maintain the steady state:

$$\omega = \alpha_3 v = 320 \text{ rad/s} \quad (12)$$

$$u(0) = \frac{T}{T_m [1 - \beta (\frac{\omega}{\omega_m} - 1)^2]} \approx 0.1042 \quad (13)$$

When $t = 0$, the tracking error $e(0) = v_d(0) - v(0)$ should be 0 since $v_d(0) = v(0)$.

Define the output value of the integrator as $x(t)$. According to the equation of PI controller (function 1) and my Simulink structure:

$$x(0) = u(0) - k_p e(0) = u(0) \approx 0.1042 \quad (14)$$

which means the initial output of the integrator in *Controller* should be 0.1042.

For the goal speed v_r :

In the subsystem *Driver Interface*, set the variable $set_decel = 1$, which means the cruise has been activated. Set the variable $ideal_v_r = 20$ at the beginning of the simulation, and when $t = 10s$, set $ideal_v_r = 30$, and thus a step increase of the goal speed has been made.

For other self-defined variables:

There are only k_p and k_i left. The adjustment of the two variables will be discussed in the following subsection.

All of the initial conditions are summarized in the table below.

Parameter	Variable	Value	Unit
Gear	$/$	3	$/$
Transmission ratio	$\alpha_n(\alpha_1)$	16	rad/m
Road slope	θ	0	deg
Initial speed (integrator initial condition)	$v(0)$	20	m/s
Goal speed	v_r	20	m/s
Proportion Coefficient	k_p	0.1	$/$
Integral Coefficient	k_i	0	$/$
Output	v	$/$	m/s

Table 4: Initial Simulation Conditions in Task 3

3.2 Adjustment of PI Control Parameters

Table 5 and figure 11 shows the adjustment of k_p and k_i of the whole system according to the system response.

Specific adjustment process:

Step 1 (Row 1 to 4): In this step, we fix $k_i = 0$ and adjust the value of k_p , and find that no matter how large k_p is, the steady-state speed cannot reach our goal speed 30 m/s, and that if k_p is too large, the throttle reaction is extremely sharp. Thus, we choose $k_p = 0.2$ where the rise time (can be regarded as reaction time) is suitable and the throttle doesn't react so sharply.

Step 2 (Row 5 to 10): In this step, we fine-tune the value of k_i . We firstly make $k_i = 0.1$ as the reference value, and notice the 2% settling time is too long, and that if we increase k_i , the settling time is even bigger, so we will decrease k_i . Finally we get $k_i = 0.006$ where the overshoot, rise time and settling time are suitable.

Step 3 (Row 11 to the end): Fine-tune the value of k_p . And thus we get the final values: $k_p = 0.24, k_i = 0.006$.

4 Task 4

There are 3 changes to the overall system:

1. Task 4 is an open-loop system (although for w it is close-loop), so we cut the u connection of the subsystem *Actuator and Throttle and Engine*, and add a constant block to represent u signal.
2. For the gear shifting simulation, we add a stateflow chart (whose input is ω and output is gear and α) to simulate the gear shifting.
3. Since α is an input of the subsystem *Gears and wheels*, we removed the constant block in this subsystem (see Figure 6).

Figure 12 shows the overall top-level architecture; figure 13 shows the stateflow chart, and figure 14 shows the new subsystem *Gears and wheels*.

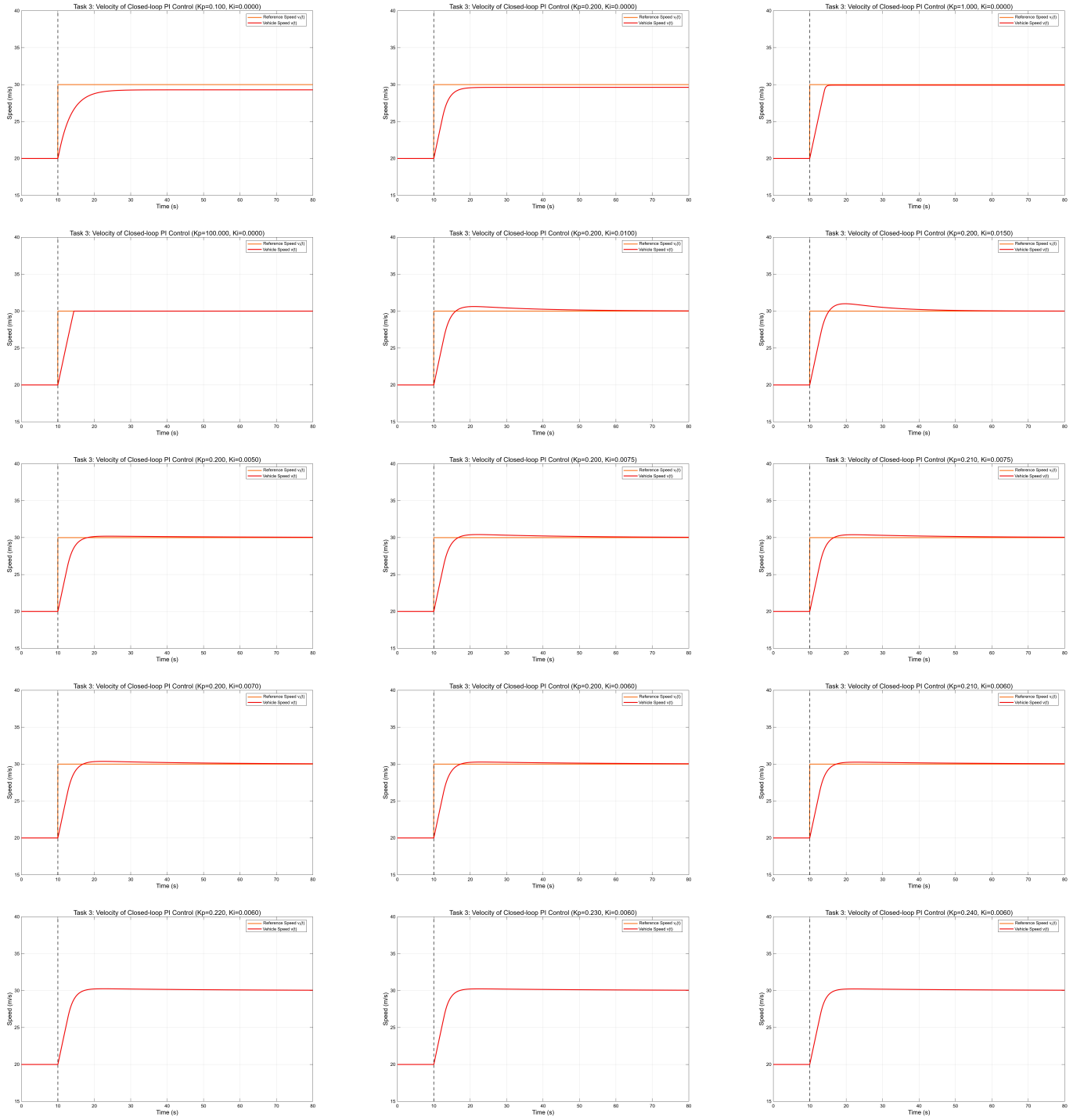


Figure 11: System Response of Different k_p and k_i Parameters (Enlarge to view)

k_p	k_i	Steady Speed (m/s)	Max Speed (m/s)	Overshoot (%)	Rise Time (s)	5% Settling Time (s)	2% Settling Time (s)
0.1	0	29.2908	29.2908	/	10.17	/	/
0.2	0	29.6290	29.6290	/	6.03	/	/
1	0	29.9229	29.9229	/	3.96	/	/
100	0	29.9992	29.9992	/	3.95	/	/
0.2	0.01	30.0125	30.6198	2.0659	4.90	4.01	12.83
0.2	0.015	30.0018	30.9999	3.3329	4.47	3.85	18.04
0.2	0.005	30.0319	30.1900	0.6318	5.74	4.26	5.63
0.2	0.0075	30.0248	30.4130	1.3768	5.24	4.12	5.23
0.2	0.007	30.0274	30.3699	1.2329	5.33	4.15	5.30
0.2	0.006	30.0315	30.2813	0.9378	5.52	4.20	5.45
0.21	0.006	30.0330	30.2654	0.8848	5.41	4.14	5.35
0.22	0.006	30.0342	30.2518	0.8394	5.32	4.09	5.26
0.23	0.006	30.0353	30.2392	0.7973	5.23	4.04	5.17
0.24	0.006	30.0363	30.2278	0.7954	5.15	4.00	5.10

Table 5: System Response of Different k_p and k_i Parameters

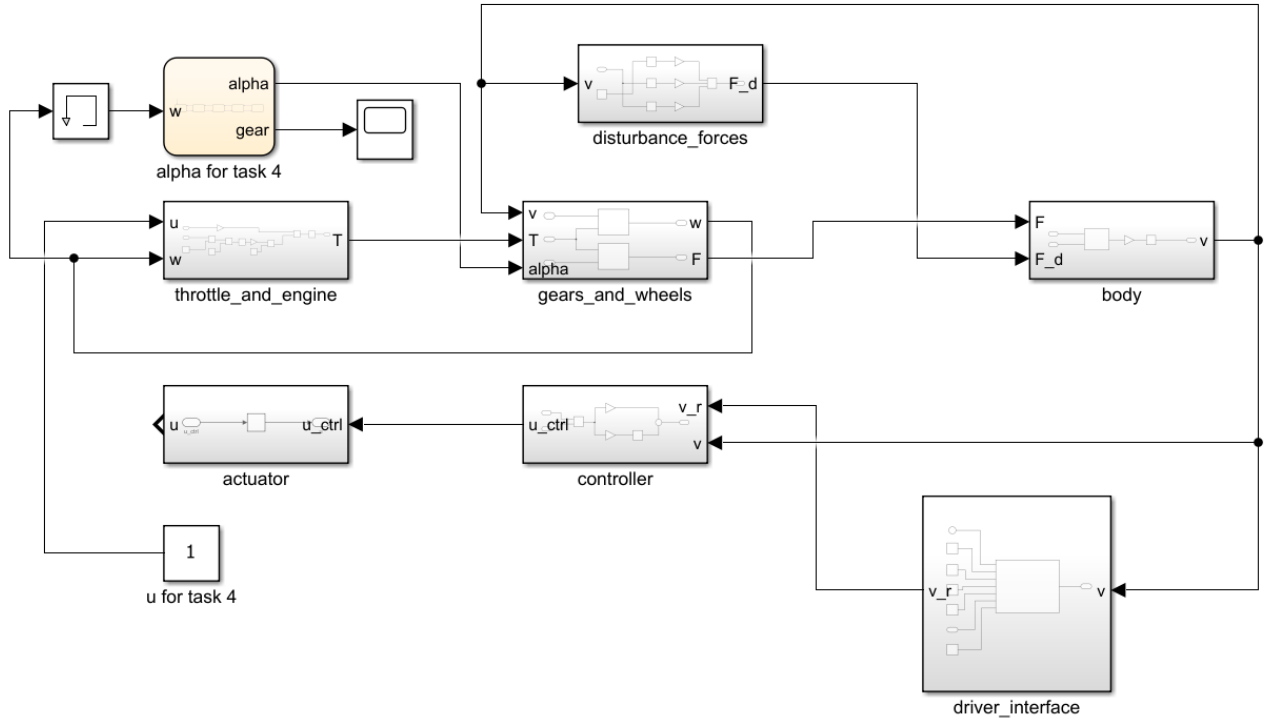


Figure 12: Overall Simulink Structure of Task 4

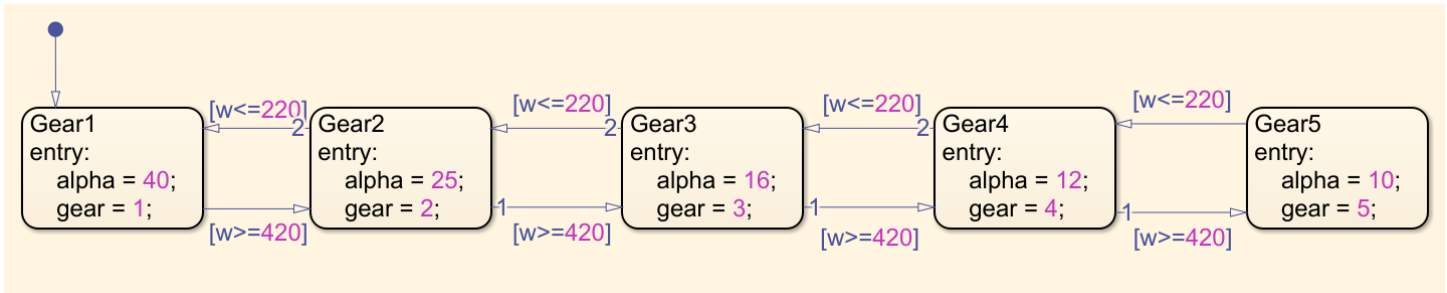


Figure 13: Stateflow Chart

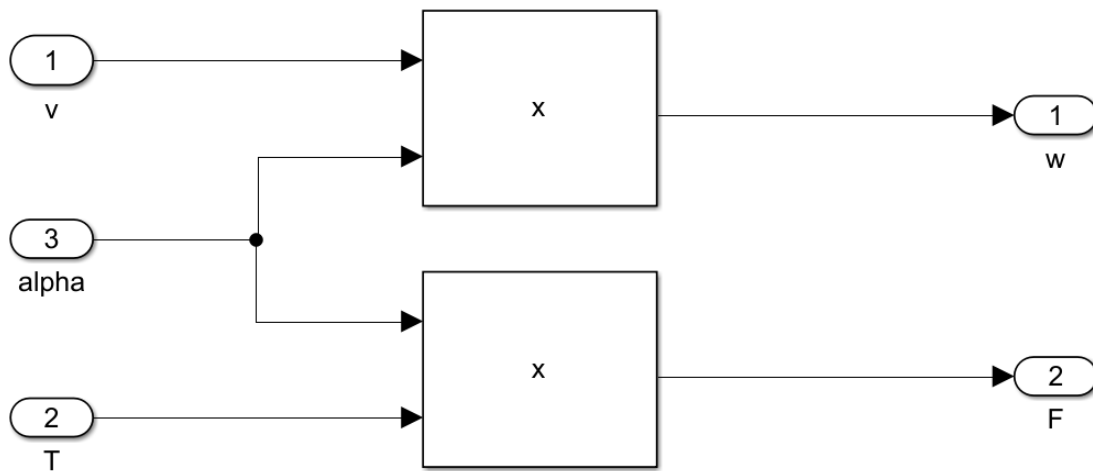


Figure 14: Structure of Subsystem *Gears and wheels* in Task 4

4.1 Stateflow Chart Design

4.1.1 Memory Block

In the top-level architecture (see Figure 12), there is a memory block before the stateflow chart. This is because an algebraic loop of ω exists in the top-level architecture, which will be refused by Simulink. To break the algebraic loop, a memory block is inserted upstream of the stateflow input. By introducing a single time-step delay, the Stateflow block receives the previous step's state instead of the current state, which eliminates the algebraic loop.

4.1.2 Transition Conditions

If the car shift the gear from i to j , since instantaneous speed remains unchanged during the transition, according to equation 4, the new engine rotation speed is:

$$\omega_j = \frac{\alpha_j}{\alpha_i} \omega_i \quad (15)$$

Define the parameter $p_i^j = \frac{\alpha_j}{\alpha_i}$, then the value of p_i^j is shown as table 6.

Gear	1-2	2-3	3-4	4-5	5-4	4-3	3-2	2-1
p_i^j	0.625	0.64	0.75	0.8333	1.6	1.5625	1.3333	1.2

Table 6: Value of parameter p_i^j when shifting gear

Define the engine rotation speed of upshifting as ω_{up} , and downshifting as ω_{down} , which should meet the following conditions preventing from gear oscillation:

$$\begin{cases} \omega_{down} \leq \min(p_i^j) \cdot \omega_{up}, & i < j \\ \omega_{up} \geq \max(p_i^j) \cdot \omega_{down}, & i > j \\ \omega_{down} \geq 0 \\ \omega_{up} \geq \omega_m \end{cases} \quad (16)$$

To ensure that the engine's performance is utilized to the fullest extent, ω_{up} should be greater but closer to ω_m as much as possible. Since $\omega_m = 450 \text{ rad/s}$, we define $\omega_{up} = 460 \text{ rad/s}$, then ω_{down} should be less than 287.5 rad/s and bigger than 0. We take the value of ω_{down} as 220 rad/s.

4.2 Simulation Conditions

The initial condition of the integrator:

- In subsystem *Body*: 0, since the initial velocity of the vehicle is 0.
- In subsystem *Controller*: any value, since the top-level system is an open-loop system.

Initial gear: 1, and at the same time $\alpha = \alpha_1 = 40$.

Table 7 shows the initial simulation conditions in task 4.

4.3 Simulation results

Figure 15 and 16 show the simulation results of vehicle speed, engine rotational speed and gear in task 4. The unit of y-axis (speed) has been changed from m/s to km/h.

According to the raw data and code execution result, the simulation results are in table 8.

When the vehicle reached 100 km/h, the gear level hasn't been fully upgraded, which is 3, not 5.

Parameter	Variable	Value	Unit
Gear	/	1	/
Transmission ratio	$\alpha_n(\alpha_1)$	40	rad/m
Throttle position	u	1	/
Road slope	θ	0	deg
Initial speed (integrator initial condition)	$v(0)$	0	m/s
Goal speed	v_r	any	m/s
Proportion Coefficient	k_p	any	/
Integral Coefficient	k_i	any	/
Output	v	/	m/s

Table 7: Initial Simulation Conditions in Task 4

Task 4: Vehicle Speed, Engine Rotational Speed and Active Gear (0-20s)

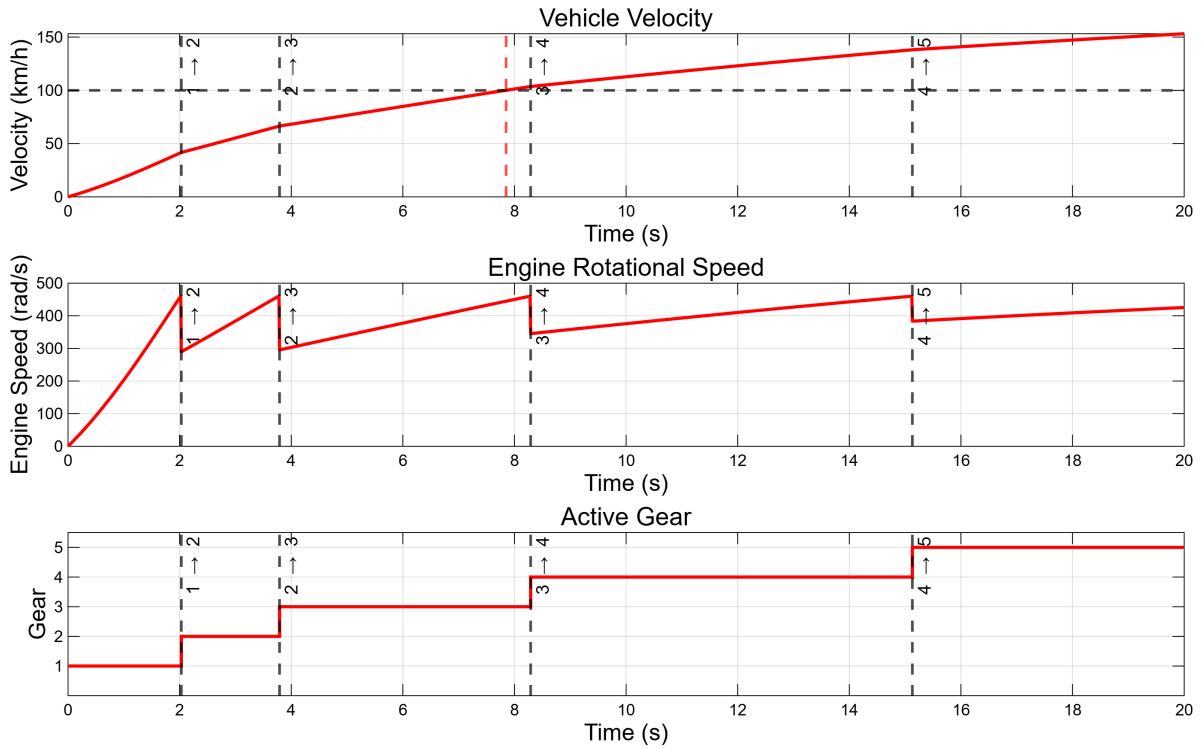


Figure 15: Simulation result of 0 - 20 s of Task 4

Time of 100 km/h	7.85 s
Time of gear 1 to 2	2.03 s
Time of gear 2 to 3	3.79 s
Time of gear 3 to 4	8.29 s
Time of gear 4 to 5	15.13 s

Table 8: Simulation Result of Task 4

Task 4: Vehicle Speed, Engine Rotational Speed and Active Gear

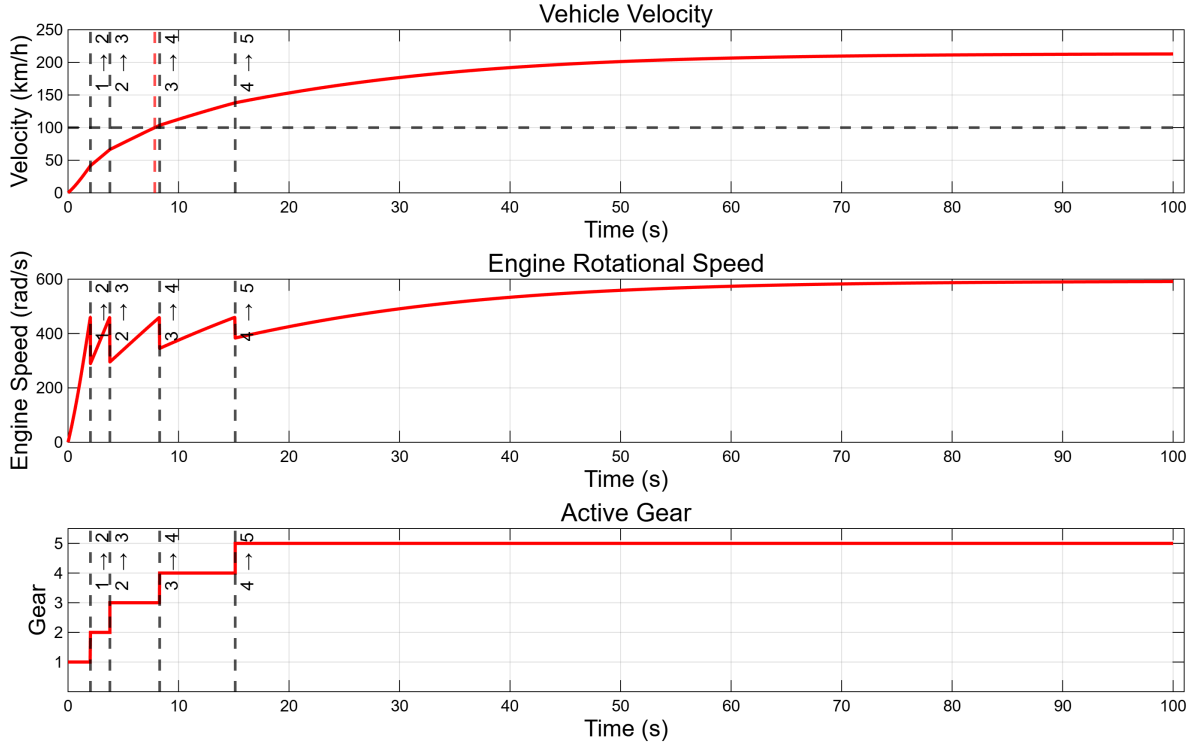


Figure 16: Simulation result of 0 - 100 s of Task 4

5 Other Information

1. MATLAB version: R2026a.

2. All the data are measured from the raw data by using the "to workspace" block in Simulink. This block will output the exact value of the measured variable and its corresponding time. No data are measured directly from the output image or the scope model from Simulink.

3. Theoretical calculation of k_p and k_i in task 3:

Perform a first-order Taylor expansion on equation 7 at the point $(u_0, v_0) = (0, 20)$ of $t = 0$:

$$m \cdot \frac{d\Delta v}{dt} \approx \frac{\partial F}{\partial u}|_{(u_0, v_0)} \cdot \Delta u + \frac{\partial F}{\partial v}|_{(u_0, v_0)} \cdot \Delta v + \frac{\partial F_d}{\partial u}|_{(u_0, v_0)} \cdot \Delta u + \frac{\partial F_d}{\partial v}|_{(u_0, v_0)} \cdot \Delta v \quad (17)$$

Since F_d is independent to u according to 6, then:

$$m \cdot \frac{d\Delta v}{dt} \approx \frac{\partial F}{\partial u}|_{(u_0, v_0)} \cdot \Delta u + \frac{\partial F}{\partial v}|_{(u_0, v_0)} \cdot \Delta v + \frac{\partial F_d}{\partial v}|_{v_0} \cdot \Delta v = \frac{\partial F}{\partial u}|_{(u_0, v_0)} \cdot \Delta u + \left(\frac{\partial F}{\partial v}|_{(u_0, v_0)} + \frac{\partial F_d}{\partial v}|_{v_0} \right) \Delta v \quad (18)$$

Define:

$$a_0 = \frac{\partial F}{\partial u}|_{(u_0, v_0)} + \frac{\partial F_d}{\partial v}|_{v_0} = \rho C_d A v_0 + 2\alpha_3^2 \beta u_0 T_m \frac{\alpha_3 v_0 - \omega_m}{\omega_m^2} \quad (19)$$

$$b_0 = \frac{\partial F}{\partial u}|_{(u_0, v_0)} = \alpha_3 T_m [1 - \beta \left(\frac{\alpha_3 v_0}{\omega_m} - 1 \right)^2] \quad (20)$$

Then we get $m \frac{d\Delta v}{dt} + a_0 \Delta v = b_0 \Delta u$. Apply the Laplace transform to both sides of the equation, and we get the transfer function $G(s)$ of $\Delta V(s)$ with respect to $\Delta U(s)$:

$$sm\Delta V(s) + a_0\Delta V(s) = b_0\Delta U(s) \quad (21)$$

$$G(s) = \frac{\Delta V(s)}{\Delta U(s)} = \frac{b_0}{ms + a_0} = \frac{\frac{b_0}{m}}{s + \frac{a_0}{m}} \approx \frac{2.5888}{s + 0.01915} \quad (22)$$

Thus, there is a pole point $s = -0.01915 \text{ rad/s}$ of the vehicle. We need to design a controller with a zero point -0.01915 to eliminate the effect of the pole, which is:

The open-loop transfer function of PI control system:

$$C(s) = k_p + \frac{k_i}{s} = \frac{1}{s} [k_p(s + \frac{k_i}{k_p})] \quad (23)$$

The open-loop transfer function of the whole system:

$$L(s) = C(s)G(s) = k_p \left(\frac{s + \frac{k_i}{k_p}}{s} \right) \left(\frac{\frac{b_0}{m}}{s + \frac{a_0}{m}} \right) \quad (24)$$

Make a pole-zero cancellation if $\frac{k_i}{k_p} = \frac{a_0}{m}$, then:

$$L(s) = \frac{k_p \cdot b_0}{m \cdot s} \quad (25)$$

The close-loop transfer function of the whole control system is:

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{\frac{k_p \cdot b_0}{m \cdot s}}{1 + \frac{k_p \cdot b_0}{m \cdot s}} = \frac{1}{\tau_{cl} + 1} \quad (26)$$

where $\tau_{cl} = \frac{m}{k_p \cdot b_0}$. If we define the time constant $\tau_{cl} = 1.5 \text{ s}$, then:

$$k_p = \frac{m}{b_0 \cdot \tau_{cl}} \approx 0.2575 \quad (27)$$

$$k_i = k_p \cdot \frac{a_0}{m} \approx 0.00493 \quad (28)$$

In task 3, we choose $k_p = 0.24$, $k_i = 0.0006$.

4. ω_m is the engine speed of maximum output torque, not the maximum engine rotational speed.

5. All the "to workspace" models have a sampling time of 0.01 s rather the default -1.